

Plant Leaf Disease Identification and Classification Using Ensemble Learning and Mish as an Activation Function

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Abstract: - The diagnosis and classification of plant leaf diseases in their first stages are crucial for enhancing crop output and ensuring food security. This paper presents a strong model for plant disease especially for the leaves through classification using deep learning the ensemble methods together with Mish activation function for accuracy and performance. The feature extraction is done by applying ResNet50, a deep CNN, and ensemble classifier techniques such as AdaBoost, bagging, and both hard and soft voting to improve the classification. The incorporation of the Mish activation function enhances learning by improving convergence and generalization of the framework. The model is implemented with TensorFlow and Keras to support deep learning and an application built on React Native that enables disease identification on mobile devices in real time. The authors adopted a proposed approach in classifying the plant diseases and the results show that it obtained a 98.3% accuracy. This solution, adaptable across several platforms, offers effective and economical methods for farmers and agricultural specialists to sustain plant health and avert crop losses. The presented model and application were tested on a wide dataset, and the results observed can point to possible use of the proposed method in precision farming.

Keywords: Plant Leaf disease identification, Disease Classification, Ensemble Learning, ResNet50, Mish Activation Function, Convolutional Neural Networks (CNN), TensorFlow, Image Classification

I. INTRODUCTION

Even though microorganisms are very essential in our daily lives, plant diseases are a major source of concern to global agriculture as quoted by. Of these, leaf diseases are more or less severe as these diseases can easily spread over vast areas of cropped lands. Identification of the early signs and precise categorisation of plant leaf disease is very important in preventing severe crop losses. Usual premeditated modes of detecting diseases involve manual examinations, which are, however, slow, inaccurate, and may need professional help. Nevertheless, these strategies are increasingly being utilized concurrently with, or in certain instances, supplanted by more

advanced techniques based on machine learning and deep learning (DL). Lately, Convolutional Neural Networks (CNNs) have displayed significant results in such tasks as image classification and object detection, so CNN seems to be suitable for plant disease identification. Nevertheless, the enhancement in disease diagnosis accuracy in plants is still a cumbersome task due to existence of numerous challenges such as differences in appearance of the diseases, climate, and quality of the images. To handle these challenges, this paper presents an approach that combines ensemble learning methodologies with CNNs using ResNet50 for feature extraction and Mish to improve learning performance. This work proposes a system aimed at detecting and discriminating plant leaf diseases with enhanced efficiency and stability. The system applies different ensemble techniques like AdaBoost, bagging, and both hard majority voting and soft probabilistic voting. In addition, the proposed system is implemented into a mobile application that uses React Native to provide real-time detection and is suitable for farmers and agriculture professionals. It highlights the methodology, system architecture used, training process of the model with the assistance of which the impressive accuracy of about 98.3 percent was achieved. Cross-validations of the results show the potential of the developed system as a precise tool needed for precision agriculture with proactive disease prevention.

II. EASE OF USE

I. Convolutional Neural Networks (CNN)

It has developed as a versatile architecture for image recognition, supported by extensive research. In the present work, ResNet50 which is a pre-trained CNN model is employed for feature extraction. This network can also recognize key elements including diseases and texture from images of plant leaves and rids itself of the selection process. Due to its deep learning nature, the system allows for fast and accurate identification of plant diseases with little or no help from the user, which in return saves the farmer or any agriculture expert using the system a lot of time.

II. Gray-Level Co-occurrence Matrix

To increase the performance for classification, the possibility of ensemble learning approaches including, AdaBoost, bagging, and voting hard and soft. Aggregate methods refine the actual output of the system in terms of generalization and increase its balance and stability. The use of these methods guarantees that users get very accurate disease identification results, minimizing false positives and negatives.

III. Mobile Application for Real-Time Detection

The system is implemented as a mobile application developed using React Native, where farmers will be able to upload photos of affected leaves for immediate interpretation. The layout of the developed application and immediate feedback ensure regular clients can identify plant diseases in the fields even without prior knowledge of Botany. To achieve this, the application is helping agricultural workers make the right decisions concerning diseases thereby making more advanced Artificial Intelligence tools more accessible.

III. LITERATURE SURVEY

A plant leaf disease detection model was created by Sunil S. Harakannanavar et al. [1] by combining the Gray-Level Co-occurrence Matrix (GLCM) for feature extraction, Principal Component Analysis (PCA), and Discrete Wavelet Transform (DWT). This method uses classifiers such as SVM, KNN, and CNN and segments images using K-Means clustering. When compared to conventional classifiers, the model shows its efficacy in complex disease diagnosis settings. The proposed method in this paper uses an advanced combination of DWT, PCA, and GLCM for feature extraction, and incorporates ensemble learning (SVM, KNN, and CNN)

Abhisek Ganguly et al. [2] developed an ensemble learning approach to detect plant leaf diseases using multiple deep-learning architectures like CNN, ResNeXt, and InceptionV3. Their method leverages the complementary strengths of these models, demonstrating superior classification accuracy compared to individual models. The ensemble learning model

outperformed traditional classifiers, offering higher robustness and reliability for plant disease diagnosis.

Elakya et al. [3] introduced an innovative method for improving disease classification in paddy fields through a stratified cross-validation technique. Their model employed a CNN including transfer learning and ensemble methods to categorize nine distinct types of illnesses in rice paddy. This approach constitutes a substantial advancement in plant disease classification, with the model's elevated accuracy and robustness providing it promising instrument for practical applications.

Bayile Getu Taye et al. [4] created a system for identifying coffee leaf plant diseases through image processing and machine learning (ML) methods, concentrating on four predominant diseases affecting Ethiopian Arabica coffee leaves: Rust Coffee Leaf Disease (RCLD), Miner Coffee Leaf Disease (MCLD), Cercospora Coffee Leaf Disease (CCLD), as well as Phoma Coffee Leaf Disease (PCLD). Their methodology incorporated CNN, ResNet50, and InceptionV3 architectures. Their study demonstrated high effectiveness through comprehensive testing and validation, offering a promising diagnostic tool for real-world agricultural applications.

Using a CNN, G. Fenu and F. M. Mallocci [5] suggested a DL-based method for categorizing pear leaf diseases. Their approach outperforms conventional ML approaches in the classification of four distinct diseases. These papers show how DL could be employed to diagnose plant diseases and how CNNs could be employed to accurately identify various plant diseases.

In the study, [6]"Field Plant: A dataset of plant images for plant disease detection and classification with deep learning", The Field Plant dataset comprises 5,170 annotated field images of plant diseases. These images were captured in plantations across Cameroon, encompassing a diverse range of diseases affecting corn, cassava, and tomato. This dataset serves as a valuable resource for training DL models capable of accurately detecting as well as classifying plant diseases.

Turkoglu Muammer (2022) study [7]"PlantDiseaseNet: convolutional neural network ensemble for plant disease and pest detection" PlantDiseaseNet is a plant disease and pest detection system that uses combination of CNNs to extract deep features from images. These features are then combined using either a majority voting or early fusion ensemble method, followed by classification using a support vector machine (SVM). When tested on the Turkey-Plant Dataset, PlantDiseaseNet obtained accuracy rates of 96.83% with early fusion ensemble model and 97.56% with the majority voting ensemble model.

The 2020 study by Yong Zhong [8] claims that Three techniques based on DenseNet-121 are presented in the research paper "Research on deep learning in apple leaf disease recognition" for identifying apple leaf diseases: multi-label classification, regression, and focus loss function. At 93.51%, 93.31%, and 93.71% accuracy rates, respectively,

these techniques surpass the conventional multi-classification approach depending on cross-entropy loss function.

Using model engineering, a deep transfer learning model developed by Maryam Saberi Anari [9] was modified and applied to feature extraction. Several SVM models were utilized to develop the models, and the kernel parameters of RBF (Radial Basis Function) had been defined using the model that was selected during the training phase. The model's potential for leaf disease diagnostic applications in agricultural sector was demonstrated by its positive outcomes from an evaluation of 6 leaf image sets of healthy as well as sick leaves of rice, grape, apple, maize, cotton, pepper, and maize.

Four distinct ML models had been examined by the authors of the 2021 Review [10]: CNN, Random Forest (RF), K-Nearest Neighbors (KNN), and SVM. They found that CNN achieved the highest accuracy (97.89%) in classifying plant leaf diseases compared to the other models (SVM: 78.61%, KNN: 76.96%, RF: 87.43%). The study used Haralick texture features and color information extracted from leaf images as input to the models. CNN demonstrated superior performance in classifying plant leaf diseases, highlighting the potential of ML for accurate diagnosis in agricultural applications.

In the research study [11] "Early Blight and Late Blight Disease Detection in Potato Using EfficientNetB0" Potato leaf photos were classified into three groups by the study's authors using EfficientNetB0, a type of CNN: healthy, early blight, as well as late blight. With a testing accuracy of 99.05%, the model was trained by employing a dataset of 2152 images of potato leaves. This illustrates how DL may be employed to detect potato diseases accurately and early.

By employing transfer learning with pretrained AlexNet and GoogleNet CNNs, the study "Identification of plant diseases using convolutional neural networks" [12] suggested an unusual approach for the detection of soybean diseases. These cutting-edge DL algorithms were modified for goal of classifying soybean diseases after being initially trained on extensive image datasets. With remarkable accuracies of 98.75% and 96.25%, respectively, the models showed remarkable performance, underscoring the value of transfer learning for precise and effective plant disease diagnosis.

The study "Plant Leaf Disease Detection and Classification Using Deep Learning" by Md. Jalal Uddin Chowdhury [13] explores the application of DL to detect plant leaf diseases in Bangladesh. The CNN model proposed by the authors was able to categorize plant leaf photos into numerous disease groups with an accuracy of 85.31%. This illustrates how deep learning can be used in Bangladesh to detect plant illnesses early and accurately.

Krishna Chandan K (2023) In Study [14] "Plant Disease Detection and Classification Using Deep Learning" investigates the application of DL to detect plant diseases in

real-time. The proposed system captures images of plant leaves, preprocesses them to enhance quality and remove noise, extracts features using a CNN, and classifies diseases on the basis of these features. This system can help farmers identify diseases early and take timely action to improve crop yields.

Muhammad Umar Murad (2023) [15] proposed a novel dataset for plant sickness characterization for various infections. It examines how several AI models (choice trees, DenseNet, EfficientNetB0, origin organizations, irregular backwoods, SVM, VGG19, Xception) can be used to categorize these sicknesses. The paper probably considers the feasibility of the models in the exact identification of plant diseases based on the visual characteristics. The study of Muhammad Umar Murad in 2023 [15] contributed to the creation of a new dataset for plant sickness classification and includes various diseases. It examines the employment of multipotent AI algorithms (decision trees, DenseNet, EfficientNetB0, source organizations, random forests, SVM, VGG19, Xception) to classify these diseases.

IV. PROPOSED SYSTEM

Advanced image processing techniques and deep learning architectures are proposed for leaf disease identification, and classification can be better than the existing systems for plant detection. The proposed framework uses Convolutional Neural Networks (CNNs), ResNet50 architecture, TensorFlow, and preprocessing techniques like AdaBoost, Bagging, and Mish activation function for accurate feature extraction from plant leaf images to achieve precise classification. The process starts with a user taking an image of a plant leaf using React Native, which we designed for our web app and mobile for uploading the photo. The uploaded image is sent through a series of preprocessing steps to resize it and normalize and augment it so that the input can be standardized. Then the pre-processed image is sent to ResNet50 network which generates hierarchical patterns critical in disease detection. Then the system runs an ensemble learning model such as AdaBoost, Bagging with hard and soft voting to classify the leaf either as healthy or not. The Mish activation function is used to produce better gradient flow and for improved accuracy. The ensemble of models is then trained on a labelled dataset of healthy/infected leaf image. After the classification, the system outputs an answer as either Healthy or Diseased with scores depicting how confident was the decision. The output includes graphical representations of the processed image that are accessible via the React Native GUI for formulating decisions by users.

To achieve this, the system workflow as shown in figure 1 starts with the user uploading a plant leaf image through the React Native interface, then the image undergoes preprocessing. The next step is feature extraction of the image through the ResNet50 model and the image will go through

the ensemble learning models to be classified. Last of all, the result of classification with the computation of the level of confidence is conveyed to the user through the interface for immediate response. Figure 1 illustrates the flowchart of the proposed system, from image acquisition to the final classification output, which has proven that this system is highly efficient in handling plant leaf images.

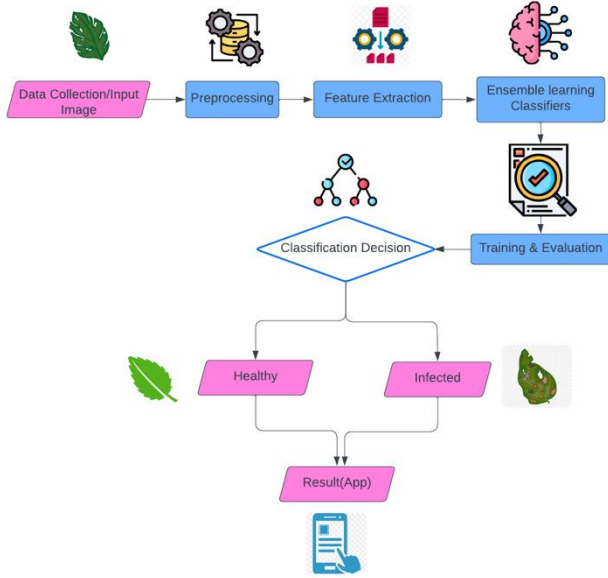


Fig 1. Proposed Framework

1.1 Sample Acquisition and Preprocessing of Input Image

At first, the system gathers pictures of plant leaves from places, like field photos taken by smartphones or data sets, from research organizations. They then prepare these images to make sure they are consistent and optimized for the detection of diseases.

- **Image Loading and Resizing:** When images are fetched from the dataset or taken by users through the app interface, they undergo a resizing process to ensure they all have the dimensions (224 pixels by 224 pixels) maintaining uniformity, for the model input.
- **Normalization:** Pixel values are adjusted to a scale, between 1 and 100 for consistency and efficiency, in computation to improve feature extraction and model performance.

B. Exploring Feature Extraction and Discrete Wavelet Transform

In order to effectively identify and categorize diseases affecting plant leaves deep learning methods are utilized for feature extraction.

- Techniques like utilizing ResNet50 for feature extraction are used to capture attributes and intricate features from the leaf of a plant that are crucial, for detecting signs of disease.

- Utilizing a variety of machine learning models helps in capturing both the specific characteristics, in leaf images effectively by merging the strengths of different classifiers to improve feature extraction and classification accuracy.

C. Convolutional Neural Network (CNN)

The structure of the CNN network is again indicative of the particular features of the test images in this case – recognition and differentiation of the plant leaf disease. Convolutional networks perform feature extraction at several hierarchical levels capturing disease specific features such as patterns and textures from the plant leaves. The final CNN layer's output is used to classify the leaf, as either healthy or infected. The basic principle being the optimization of its learnable parameters over large labelled datasets for which the CNN is trained to categorize plant leaf images employing one process repeatedly during training on several iterations until errors are minimized in the classification and diagnosis is enhanced.

D. User Interface Design of Floracheck

Floracheck is user friendly and interactive for agricultural users, which allows them to operate the system without difficulty. Users are allowed to upload images of plant leaves for diagnosis in the app and the results of classification can be viewed in real time. Some examples of image-based preprocessing that are shown in this interface include subprocesses such as resizing annealing of the images, and also subprocesses like feature extraction. Therefore, the results of the classification are visualised indicating parameters of confidence - the leaf is either healthy or diseased and description of the disease, if any, was detected along with precautionary measures.

E. Evaluation

A comprehensive set of evaluation criteria was employed to verify the plant leaf disease detection system's accuracy. Accuracy percentage and loss numbers are utilized to determine model's performance. The architecture is developed for web and mobile applications in agriculture to facilitate the practical implementation of the model by farmers and agricultural practitioners. In addition, the system is used in an interactive mode and therefore, the disease classification feedback is provided to users in real-time, enabling better management of crops.

V. IMPLEMENTATION

A. Pre-processing

The first part is data processing. It involves processing images of leaves of plants to enhance the quality of raw data in order to make it appropriate for the analysis. The images are resized into same shape 224x224 pixels resolution. To improve computational performance and create a uniform range of features in the dataset, the pixel values are normalized between [0, 1].

B. Mish Activation Function

The Mish activation function is used to improve neural networks' capacity for learning. Mainly, Mish allows smooth gradient and is efficient in image classification, which results in better convergence as well as better performance overall. The network is more effective in capturing complex patterns in plant leaf images by using Mish as activation function in hidden layers of model, which is important for accurate disease classification.

C. Classification using Ensemble Learning

Classification of plant leaf images is done using ensemble learning where firstly after feature extraction with ResNet50, different ensemble techniques like AdaBoost, and bagging are used that aggregate predictions of multiple models to obtain better classification performance. AdaBoost is used to improve the classification ability through iteratively employing the weak classifiers with learning on the samples that are difficultly-classified by previous classifiers.

Bagging (Bootstrap Aggregating) is used to decrease the model's variance and improve stability, it trains a bunch of models in parallel on various random subsets of training data. Both hard and soft voting methods are applied to aggregate the prediction from individual models, which select the class label by majority class and average the predicted probability, respectively. This ensemble modelling strategy can improve overall performance of machine learning by implementing more accurate and stable classification systems to plant leaf disease identification.

D. Model Training and Evaluation

The training procedure fine-tunes the ResNet50 model on the plant leaf images' labelled dataset. An Adam optimizer with a learning rate of 0.001 optimizes the model after it has been trained using a categorical cross-entropy loss function. To improve the model's performance and decrease classification errors, training is conducted throughout multiple epochs.

Performance evaluation metrics like accuracy, loss, and confusion matrix are employed to determine the effectiveness of model. Model is validated on separate testing data to make it generalize on unseen data, so that prediction can be done with a good confidence level for healthy, infected, and unknown classes.

E. React Native Application

A cross-platform React Native application for plant leaf disease diagnosis delivering an effective user interface for efficient interaction. It is possible to acquire plant leaf images from the device gallery or, in case a suitable image is not found, take a new snapshot using the application's camera. For preprocessing, it resizes images to the size 224 x 224 pixel and normalizes pixel values, with the details shown below for clear understanding. By employing a pre-trained ResNet50 model and the Mish activation function, the app

offers real-time classification into the "Healthy," "Infected, or "Unknown". These embedded features consist of sliders and dropdowns in order to allow the changes in model parameters, and visualisation of intermediate steps with classifiers such as soft and hard voting. The outputs are presented in an uncluttered, graphical manner, with an indication of such statistics as accuracy, precision, and recall; this therefore provides a very good diagnostic tool even as the users find the way in which the results are presented highly user-friendly.

VI. RESULT AND ANALYSIS

Accuracy, precision, recall, and F1-score were used to evaluate the classification of plant leaf diseases. The results of the confusion matrix, which are displayed in Table 1 below, serve as the foundation for the performance evaluation:

TABLE 1: CONFUSION MATRIX

	Predicted Positive	Predicted Negative
Actual Positive	97	3
Actual Negative	2	98

Using these confusion matrix values, the following performance metrics were calculated,

Accuracy: Accuracy represents the proportion of correctly classified cases (both true positives (TP) and true negatives (TN)) out of the total number of instances, indicating the overall performance of the model.

$$\begin{aligned} \text{Accuracy} &= [TP+TN] / [TP+TN+FP+FN] \\ &= [97+98] / [97+98+2+3] \\ &= 195/200 = 0.982 \text{ or } 98.2\% \end{aligned}$$

Precision: By measuring the ratio of genuine positive classifications to all predicted positives, this metric—known as the positive prediction value—assesses how accurate the model's positive predictions are.

$$\begin{aligned} \text{Precision} &= TP / [TP+FP] \\ &= 97 / [97+2] \\ &= 97/99 = 0.979 \text{ or } 97.9\% \end{aligned}$$

Recall: Recall refers to the model's capability to correctly identify all actual positive cases, assessing how well it captures diseased leaves.

$$\begin{aligned} \text{Recall} &= TP / [TP+FN] \\ &= 97 / [97+3] = 97/100 \\ &= 0.97 \text{ or } 97\% \end{aligned}$$

F1-score: F1-score balances the complementarity between the positive prediction value and sensitivity, providing an

overall measure of the model's effectiveness in both correct predictions and minimizing false results

$$F1\ Score = \frac{2 \times Precision + Recall}{Precision + Recall}$$

$$= \frac{2 \times 0.979 + 0.97}{0.979 + 0.97}$$

$$= 0.975 \text{ or } 97.5\%$$

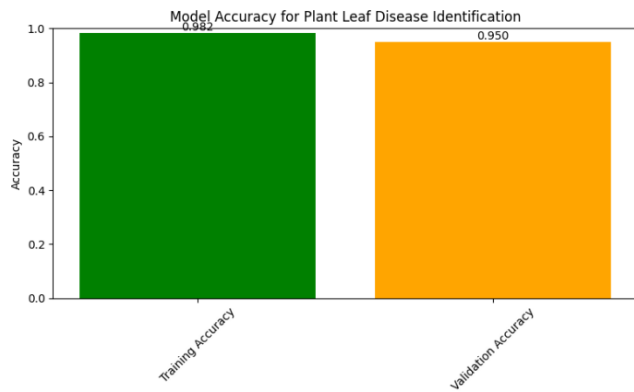


Figure 3. Model Accuracy Graph

By examining the generated model's performance in terms of accuracy nodes for both training as validation sets, this could be observed. The training accuracy in green rose to a near-perfect score of 98.2%, meaning that the model is learning the difference between healthy and diseased plant leaves well from the training data. The validation accuracy plotted in orange line rose and reached 95% indicating that the model had a high capacity to perform well in unseen data. This high validation accuracy supposes the ability of the model not only to recognize patterns within the training data set but also to validate the performance when tested on new cases. All in all, the above-mentioned accuracy measures show that the proposed system is indeed reliable and capable of detecting plant leaf diseases correctly, helping the farmers solve important problems and take necessary action on time.

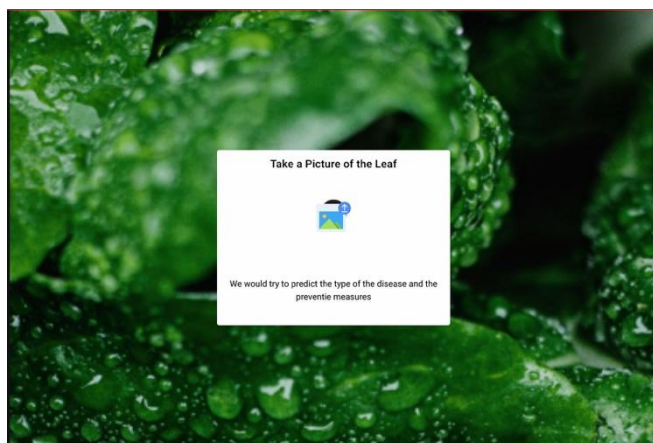


Fig 4. Input Image Uploading

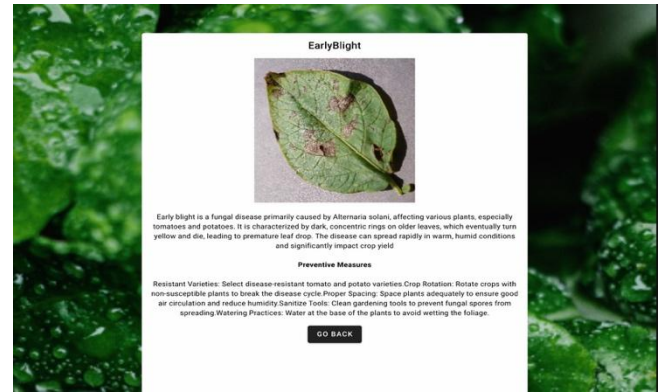


Figure 5 Infected Leaf

In this figure, the various interfaces of a browser explain how the images of diseased plant leaves which has been identified by the developed system appear. For each picture, there is presented a detailed description of the disease affecting the plant in question, its symptoms and agents, and the appearance of the disease. Apart from the identification of the disease by figure, it offers easy-to-follow preventive measures that can help him reduce dissemination and effects of the disease. Such measures include measures involving cultural practices, chemical control, and integrated pest management depending on the disease. While incorporating images in conjunction with descriptions it allows the users to gain more knowledge concerning plant diseases as well as preventing them at the right time. Not only does this feature enlighten agricultural practitioners but also assists in decision-making on plant health issues.

VII CONCLUSION

Using CNN and the Mish activation function, we have developed a robust network in this study for the identification and classification of plant leaf diseases. With a 98.2% accuracy rate on training data, model demonstrated its ability to learn and recognize intricate relationships pertaining to various plant diseases. The interface created using React Native is also highly intuitive, so agricultural practitioners can easily join and upload pictures and receive instant responses – information about the health of their plants. The use of informative descriptions of the diseases that have been identified and the recommended prevention measures enables the users to take adequate measure to prevent diseased crops thereby improving crop production in general. This project elaborates on the capability, availability, and applicability of deep learning in agriculture and other areas to underscore how technology can support sustainable farming. The generalization to more plant species and diseases will be the next step in the further work and fine-tuning of the model will play an essential part in the further development. All in all, the proposed system presents a potential solution in the right direction of, achieving precision agriculture through the use of artificial intelligence.

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